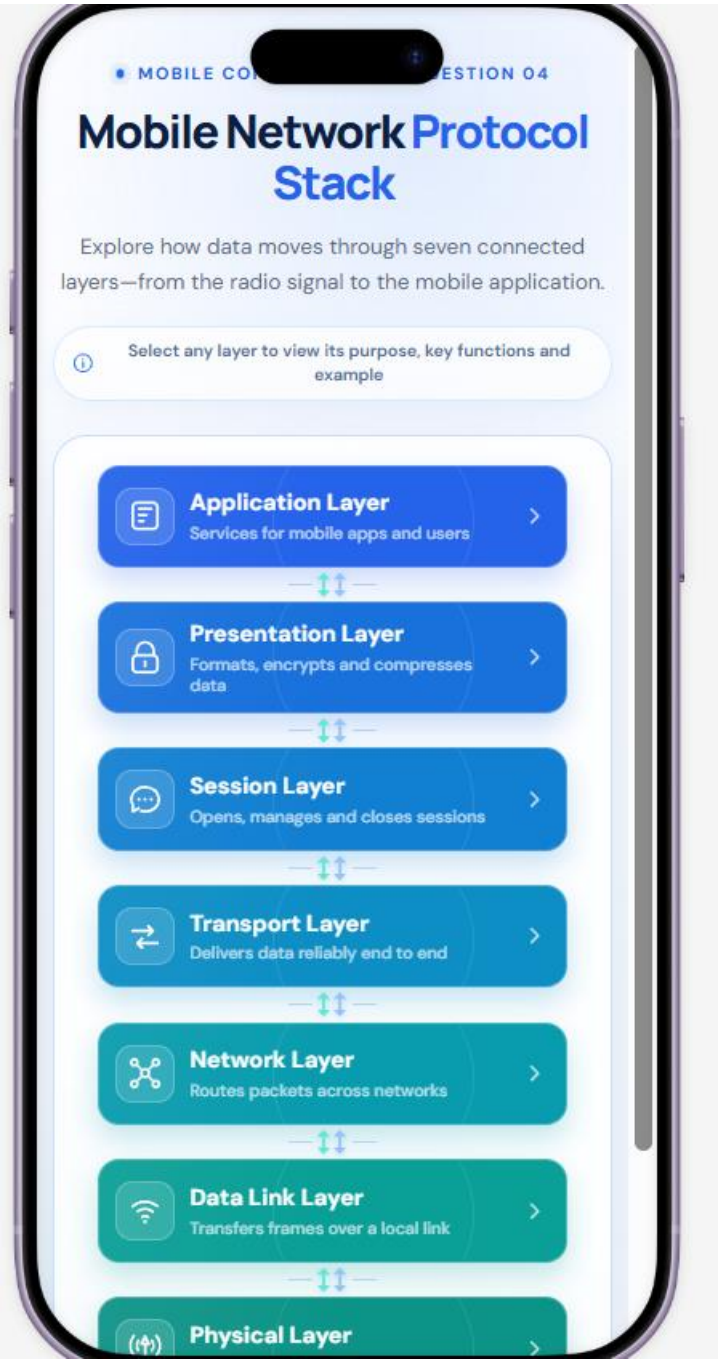
4. Create a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

Aim: -
Design a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

Procedure: -
1. open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
11. Review and edit the Prototype
12. Save and Share

Design: -


MOBILE CO ESTION 04
Mobile Network Protocol Stack
Explore how data moves through seven connected layers—from the radio signal to the mobile application.
Select any layer to view its purpose, key functions and example
Application Layer
Services for mobile apps and users
Presentation Layer
Formats, encrypts and compresses data
Session Layer
Opens, manages and closes sessions
Transport Layer
Delivers data reliably end to end
Network Layer
Routes packets across networks
Data Link Layer
Transfers frames over a local link
Physical Layer

LAYER 7

Application Layer

PURPOSE

Provides the network services that mobile applications use to communicate with remote systems.

KEY FUNCTIONS

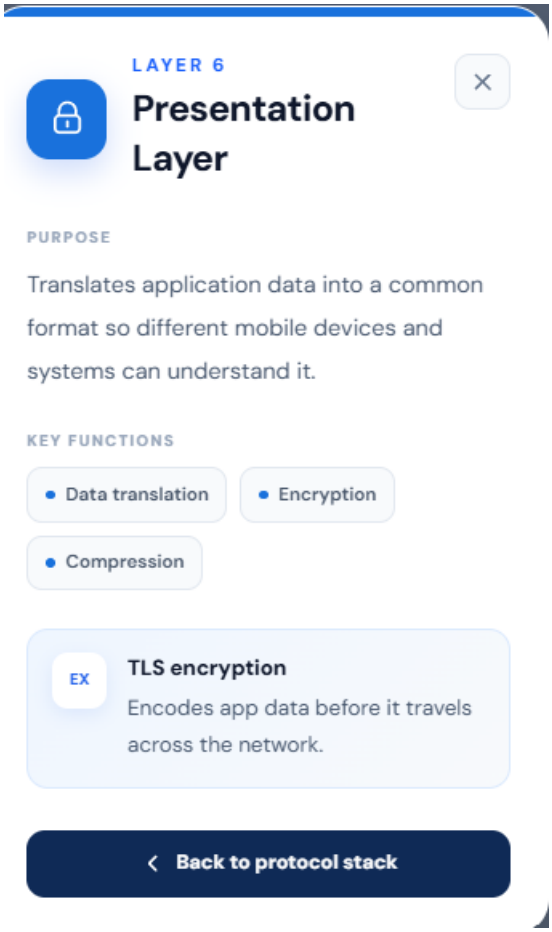
- Web access
- Email services
- Name resolution

EX

HTTPS

Securely loads content in a mobile browser or app.

Back to protocol stack


LAYER 6
Presentation Layer
PURPOSE
Translates application data into a common format so different mobile devices and systems can understand it.
KEY FUNCTIONS
Data translation
Encryption
Compression
EX
TLS encryption
Encodes app data before it travels across the network.
Back to protocol stack

LAYER 5

Session Layer

PURPOSE

Coordinates communication sessions between mobile applications and keeps their conversations organized.

KEY FUNCTIONS

- Session setup
- Synchronization
- Session recovery

EX

SIP session

Establishes and manages a voice or video call session.

Back to protocol stack

LAYER 4

Transport Layer

PURPOSE

Moves data between endpoints, controlling reliability, flow, and the rate of transmission.

KEY FUNCTIONS

- Segmentation
- Error recovery
- Flow control

EX

TCP / UDP

TCP favors reliability; UDP favors speed for streaming.

Back to protocol stack

LAYER 3

Network Layer

PURPOSE

Selects paths and addresses packets so data can travel between the mobile device and distant networks.

KEY FUNCTIONS

- Logical addressing
- Packet routing
- Path selection

EX

IPv6

Provides an address and route for each packet across the internet.

Back to protocol stack

LAYER 2

Data Link Layer

PURPOSE

Organizes bits into frames and controls reliable access to the immediate wireless or wired medium.

KEY FUNCTIONS

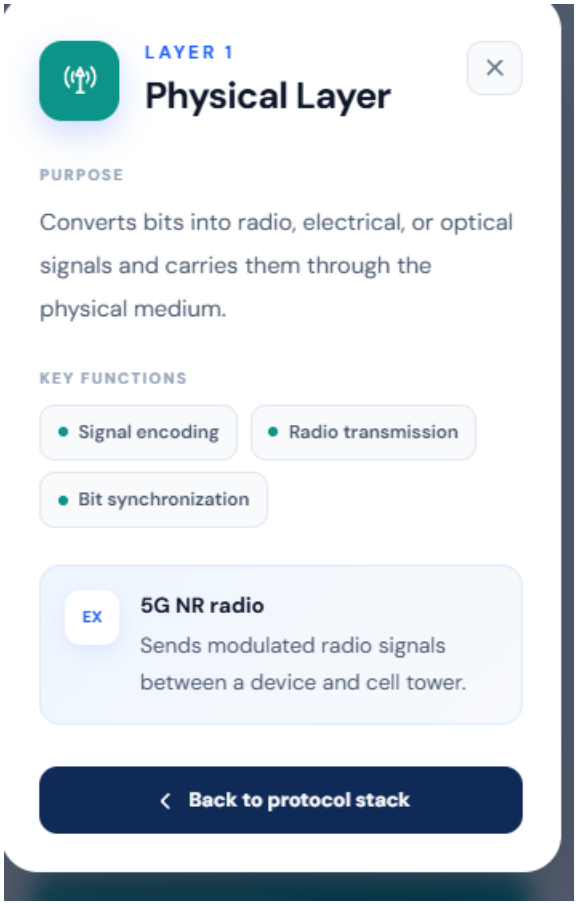
- Framing
- MAC addressing
- Link error detection

EX

Wi-Fi (802.11)

Moves frames between a phone and its wireless access point.

Back to protocol stack


LAYER 1
Physical Layer
PURPOSE
Converts bits into radio, electrical, or optical signals and carries them through the physical medium.
KEY FUNCTIONS
Signal encoding
Radio transmission
Bit synchronization
EX
5G NR radio
Sends modulated radio signals between a device and cell tower.
Back to protocol stack